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Next Generation AI Researchers for Air Quality Excellence



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Introduction to Big data



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What is traditional data?

- Structured, relational data that organizations have been using for decades
- Used for:
 - Sales tracking
 - Customer relationship management (CRM)
 - Workflows
- Advantages include easier manipulation and management of conventional software, but limited insights compared to Big Data



What is BIG DATA?

- 'Big Data' is similar to 'small data', it just differs in size
- But using big data requires a different approach:
 - Techniques, tools and architecture
- Big data presents challenges in storing and processing very large amounts of data that cannot be addressed by traditional techniques



What is BIG DATA?

- Walmark receives transactions from more than 1 million customers every hour
- Facebook processes 40 trillion images from its users
- It used to take 10 years to decode the human genome, now it takes a week
- **2010:** The world created 1.2 zettabytes of data
- **2025:** It is predicted to grow to 175 zettabytes or more



The foundations of the Big Data revolution

- Data sets too large and complex for traditional relational databases (RDBMS).
- Historical Context: Born out of Google and Yahoo's need to index billions of pages in milliseconds.
- Transition from a data-poor environment to an information-rich environment.



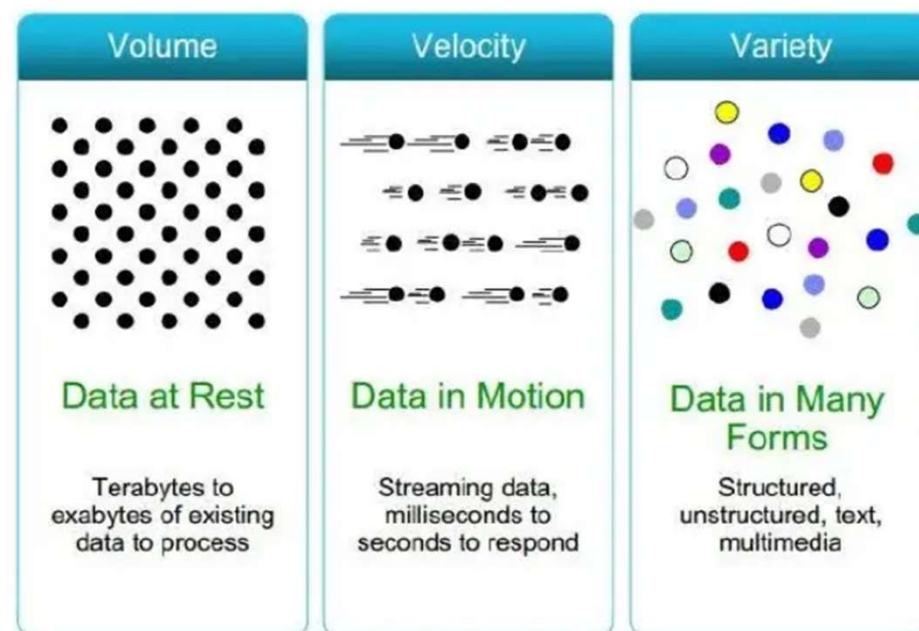
What is Big Data? – a practical definition

- Data that exceeds traditional systems in size, velocity, or complexity
- Big Data refers to vast amounts of information characterized by constant expansion in the digital universe
- Includes structured, semi-structured, and unstructured data
- Requires distributed processing and storage
- Focuses on analytics, not just reporting



Characterization of Big Data – V3

- Four key dimensions that define Big Data:



Volume

- **Scale and size of data generated and stored**
- Large amounts of data are continuously generated (smart meters, sensors, logs)
- Data is growing faster than storage costs are decreasing (Exponential growth from terabytes to **petabytes and beyond**)
- Example:
 - In 2000, a typical PC had 10GB of memory
 - Today, Facebook processes 500 terabytes of new data every day
 - A Boeing 737 generates 240 terabytes of data during a single flight



Velocity

- Data is arriving continuously and rapidly (data is generated, processed and analyzed in real or near real time), some decisions must be made in a matter of seconds
- Bundled processing is often too slow
- Example:
 - Stock exchange algorithms display changes in the stock market every microsecond
 - Sensors generate large amounts of log data in real time
 - Online gaming systems support millions of users, who produce multiple data sets every second
 - Social media feeds



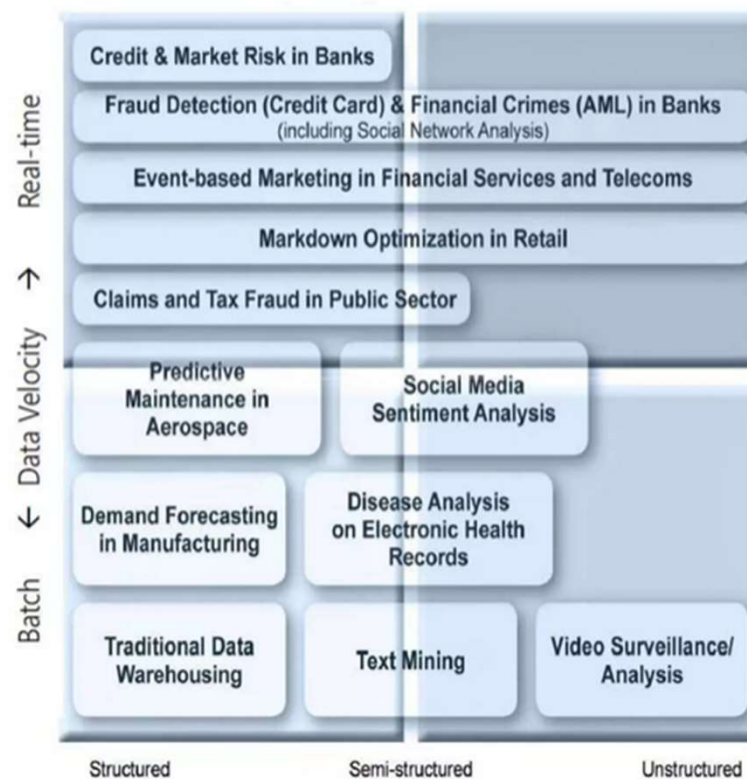
Variety

- Different types of data formats and sources in the ecosystem
- Big Data is not just numbers, dates and strings. Big Data also includes geospatial data, 3D data, audio and video recordings, and unstructured text, including log files and social networks.
- Traditional database systems are designed to process smaller amounts of structured data, with fewer changes or with a predictable and consistent data structure.
- Big Data data analysis includes different types of data: **structured**, **semi-structured** and **unstructured data**.
- Structured data: ERP tables
- Semi-structured data: JSON, logs
- Unstructured data: documents, images



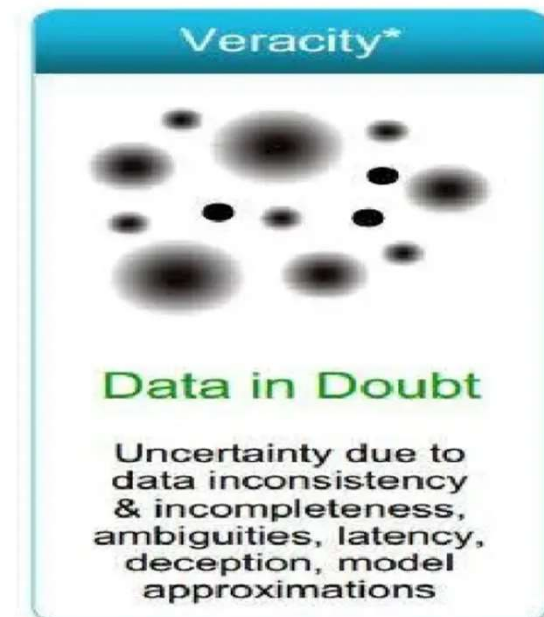
Big Data Structure

- Structured data
 - Traditional data sources
- Semi-structured data
 - Most data sources for big data
- Unstructured data
 - Video and audio data



Veracity – 4V

- It implies reliability, accuracy and credibility of data
- Problems arise:
 - Uncertainty
 - Noise
 - Bias within large data sets
- Need to perform:
 - Data cleaning
 - Detect anomalies
 - Ensure integrity



Today 5V Model – Value

- Not all data is valuable
- Value depends on the business use case
- Analytics must support decisions



Key differences: Size, organization, sources and methods of analysis

- Traditional data is measured in gigabytes and terabytes, while Big Data reaches **petabytes, zettabytes** and **exabytes**
- **Data organization:**
 - **Traditional:** Fixed schema, relational databases (SQL, Oracle, MySQL)
 - **Big Data:** Dynamic schema, raw and unstructured data, NoSQL databases like **Cassandra** and **MongoDB**
- **Sources:**
 - **Traditional:** ERP, CRM and online transactions
 - **Big Data:** Social networks, device sensors, audio and video recordings
- **Analysis methods:**
 - Traditional analysis happens **after the event** (incremental)
 - Big Data analysis enables **real-time** insights as data is collected



BI vs. Big Data: A Paradigm Shift

- Traditional BI: "Schema-on-Write" – data is cleaned and structured before storage in the Warehouse.
- Big Data approach: "Schema-on-Read" – storage in raw form (Data Lake), and the structure is added upon request.
- Objective: BI is for repeatable reports; Big Data is for research analytics and machine learning.



Why classic technologies are not enough

- Relational databases have scaling limitations
- Vertical scaling is expensive
- The data structure must be predefined
- Working with IoT and real-time data is difficult
- Example:
 - "If we have thousands of sensors on a network, the classic base becomes a bottleneck."



Hadoop – the foundation of the Big Data ecosystem

- Apache Hadoop is a platform for:
 - distributed data storage
 - large data processing
- Hadoop is not a single technology, but an ecosystem.



Major components of Hadoop

- **HDFS (Hadoop Distributed File System)**
 - Data is divided into blocks
 - They are preparing for multiple servers
 - Resistant to failures
- **YARN**
 - Resource Management
 - Distribution of work between nodes
- **MapReduce**
 - A model for distributed data processing
 - Map phase: data processing
 - Reduce phase: aggregation of results
- **Other tools in the ecosystem:**
 - **Hive** – SQL-like queries over HDFS
 - **Pig** – a scripting language for data processing
 - **HBase** – NoSQL database on top of HDFS
 - **Sqoop** – data transfer between RDBMS and Hadoop
 - **Flume** – collecting log data



When is Hadoop used? / Limitations of Hadoop

- Long-term storage of large amounts of data
- Low-cost storage of historical data
- Logs, sensor data, archives
- Slower for real-time processing
- More complex to maintain
- Today it is often used in combination with the cloud



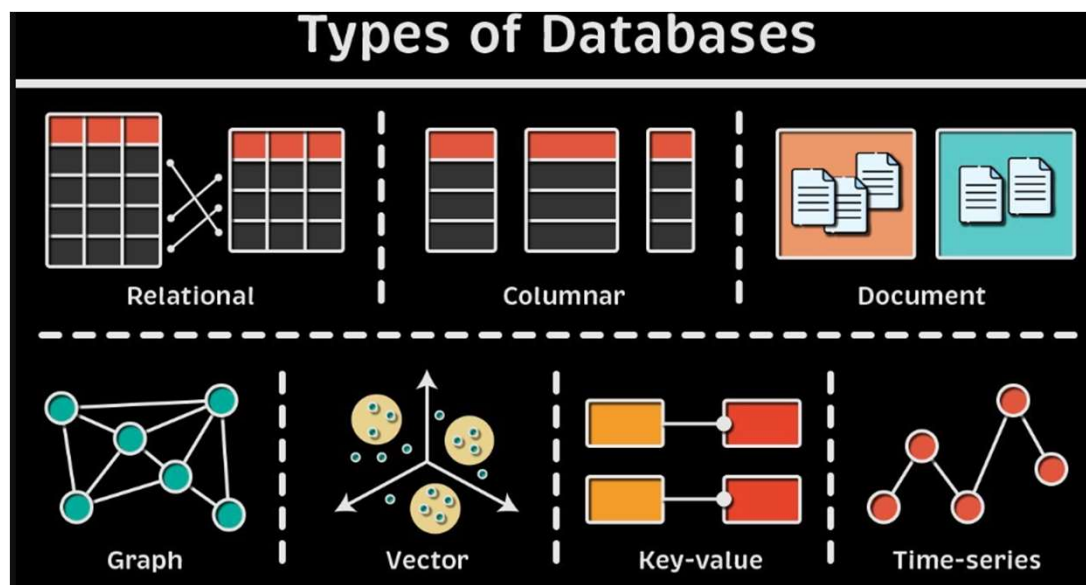
NoSQL databases

- NoSQL databases are created because:
 - data does not always have a fixed structure
 - a large number of records are required in a short time
 - scaling must be easy



Types of NoSQL databases

- **Key-Value base**
 - very fast
 - simple structure
- **Document databases (e.g. MongoDB)**
 - flex JSON records
 - often used in applications
- **Column databases (eg Cassandra)**
 - large amounts of data
 - high availability
- **Time-series bases**
 - sensor and weather data
 - energy and IoT
- **Graph bases (eg Neo4j)**
 - Relationships between data in focus



Cloud platforms for Big Data

- **What does "Big Data in the cloud" mean?**
- Cloud platforms offer:
 - ready-made Big Data services
 - scaling without buying infrastructure
 - payment according to use



Advantages / Challenges

- **Advantages of the cloud**
 - Quick launch of projects
 - Lower initial cost
 - Easier experimentation (pilot projects)
- **Cloud challenges**
 - Safety
 - Regulation
 - Integration with existing systems

Technology	Purpose	Advantages	Disadvantages
Hadoop	Batch processing	Stable, scalable	Slower
Spark	Batch + Stream	Fast, flexible	Higher memory consumption
NoSQL	Storage	Scalability	Weaker consistency
Cloud	Infrastructure	Flexibility	Cost



The most common cloud platforms

- **Microsoft Azure**
 - often used in industry
 - strong integration with existing IT systems
- **AWS (Amazon Web Services)**
 - very rich ecosystem
 - flexibility and scalability
- **Google Cloud (GCP)**
 - strong analytics
 - High processing speeds
 - BigQuery



Google Cloud



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What is most important to remember

- Hadoop = big data storage
- Spark = analytics and processing
- NoSQL = flexible databases for high speeds
- Cloud = scalability and speed of implementation



Big Data architecture



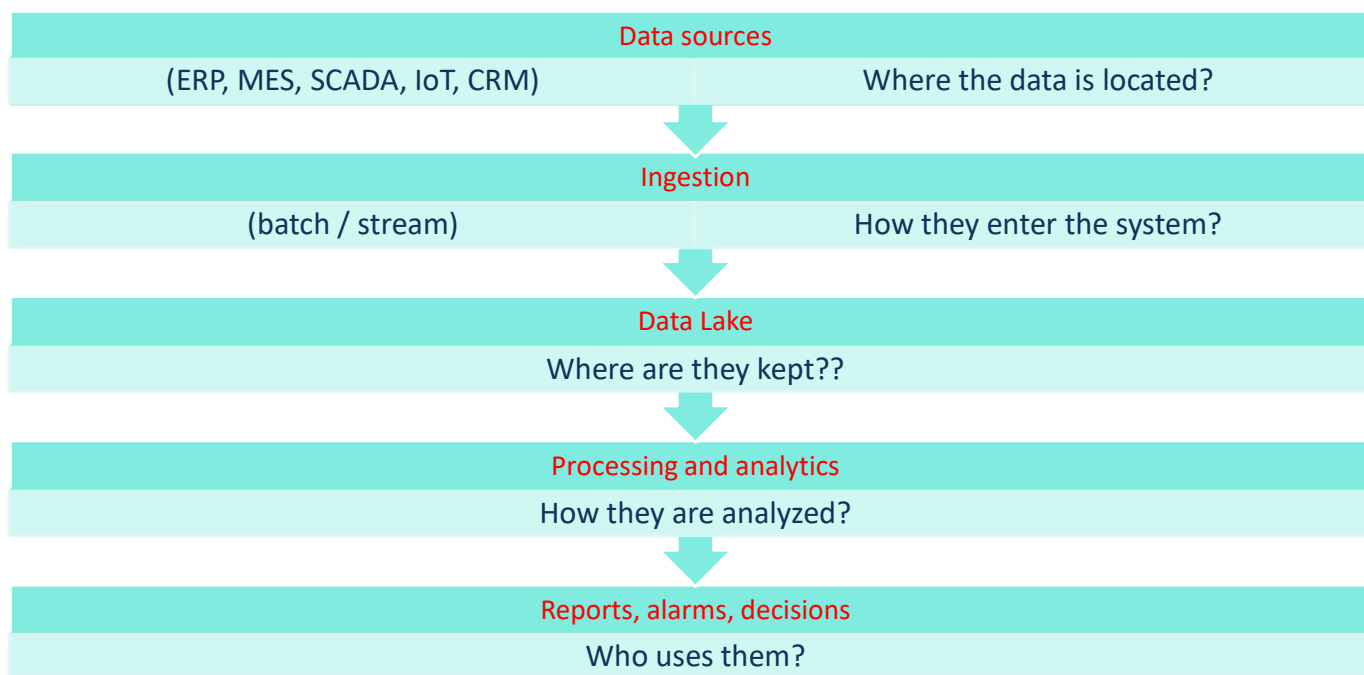
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Typical challenges without a Big Data architecture

- Data in silos
- Manual transfers (Excel)
- Information delay
- Unreliable reports



Basic data flow in Big Data system

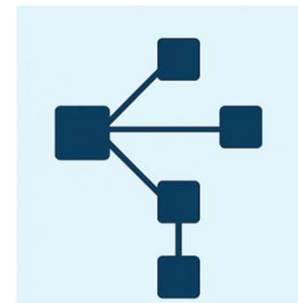
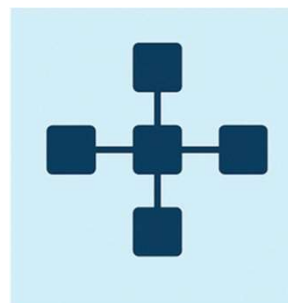


Data Warehouse – traditional approach

- **What is a Data Warehouse?**

- Centralized base:

- structured data
- predefined model
- optimized for reporting



- Data is stored in a relational database and queried using the SQL query engine
- Tables are denormalized to optimize queries
 - Usually as a star or snowflake schema of numerical facts that can be aggregated by dimensions

Characteristics of Data Warehouse / When to use Data Warehouse?

- The scheme is predefined
- Data is cleaned before storage
- Stable reports
- Less flexible
- Financial reporting
- Statutory reports
- Standard KPIs
- It stores highly structured, processed data optimized for business reporting (BI) and fixed queries.



Data Lake – Big Data approach

- **What is a Data Lake?**
- Central Repository:
 - all types of data
 - structured and unstructured
 - stored in their original form



Data Lake characteristics

- "Schema-on-read"
 - Large amounts of data
 - Flexibility
 - Suitable for analytics and AI
-
- Data files are stored in a distributed file system (a data lake) and are typically processed using Apache Spark
 - Metadata is used to define tables that provide a relational SQL interface to file data
 - Typically, the data lightweight format is used to provide transactional database functionality



What data goes into the Data Lake?

- IoT sensors
- Machine logs
- SCADA data
- External data (time, market)
- Stores raw data in native format (structured and unstructured), **ideal for data mining and machine learning.**
- **Data Lake Risks**
 - Bad management → "data swamp"
 - Discipline and governance required



Data Lake vs. Data Warehouse

Data Warehouse	Data Lake
Structured data	All data types
Strict rules	Flexibility
Reporting	Analytics & AI
Less scalable	Highly scalable



Data Lakehouse

- A hybrid approach that combines the low cost and flexibility of a data lake with the performance and reliability of a data warehouse

Feature	Data Warehouse	Data Lake	Data Lakehouse
Data structure	Highly structured	Raw, various formats	Raw with metadata
Schema	Schema when writing (on-write)	Schema during reading (on-read)	Flexible scheme
Users	Business analysts	Data scientists	Both groups
Storage cost	Higher	Lower	Economic balance



Big Data data flow



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Batch data processing

- Data processing in time intervals
 - Data is collected → then processed
 - Collects, stores and processes data **at scheduled intervals**
(e.g. at night or off-peak)
- Examples:
 - Daily reports
 - Production planning
 - Analysis of historical data



Advantages and limitations

- Simpler
- Stable
- **There is no real-time reaction**



Stream data processing

- **What is stream processing?**

- Data is processed as soon as it is created
- Minimal delay
- It processes and analyzes data as it is received, enabling immediate insights and actions

- Examples of stream processing:
 - Alarming
 - Anomaly detection
 - Real-time network monitoring
 - IMROH Information system for monitoring automatic stations for air quality



Batch vs. Stream – when what?

- **Batch** = planning and analysis
- **Stream** = monitoring and reaction

Feature	Batch Processing	Stream Processing
Latency	High (hours or days)	Low (milliseconds/seconds)
Data handling	Data is first stored and then processed	Data is transformed as it is generated
Complexity	Easier to implement	More complex architecture
Amount of data	Veliki blokovi podataka	Continuous flow of data



Defining Big Data projects



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Why most Big Data projects fail

- Most Big Data projects fail not because of technology, but because of:
 - poorly defined problem
 - unrealistic expectations
 - lack of business ownership
- Typical reasons for failure
 - "We do Big Data because everyone does"
 - Unclear goal
 - There is no owner
 - There is no plan for production



How to identify the right Big Data use-case

- **Starting point: What is problem???**
- A good use case always starts with a question:
 - What do we want to improve?
 - What problem do we want to solve?
 - Where do we waste money, time or quality?
- Characteristics of a good use-case
 - Measurable business value
 - Available or realistically available data
 - Clear beneficiaries of the results
 - Limited scope (pilot)



Resource and technology assessment

- **What resources are needed?**

- People
- Time
- Infrastructure
- Budget

- **Common mistakes:**

- The data exists "somewhere"
- The data is not integrated
- The data is not reliable

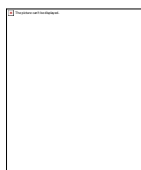
- **Technological assessment:**

- Existing IT infrastructure
- Cloud vs on-premise
- Security requirements





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Thank You!

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